

RegNetAgents Multi-Agent Architecture

(LangGraph-Orchestrated Workflow)

Input

User Query via MCP Server
(Natural Language)

Initialization & Validation Agent
Gene Symbol → Ensembl ID

ARACNe Networks
(Pre-computed
NetworkX cache)

Network Modeling Agent

- Identify Regulators (upstream)
- Identify Targets (downstream)
- Classify Regulatory Role

Analysis

>5
regulators?

Therapeutic Target
Prioritization

- PageRank
- Centrality
- Target
Ranking

Pathway Enrichment Agent
Reactome API Query

→ Reactome
Database

Parallel Domain Analysis (LLM-Powered)

Domain
Insights

Cancer
Biology
Agent

Drug
Discovery
Agent

Clinical
Relevance
Agent

Systems
Biology
Agent

Integration & Reporting Agent

- Synthesize multi-domain insights
- Generate comprehensive JSON report
- Rank therapeutic targets

Output

Comprehensive Analysis Report
(Structured JSON)

Execution Time:
~0.6s (rule-based)
~15s (LLM-powered)

Legend:



Required step



Conditional execution